

★ WEEK 14 · PART 2 of 3

Metropolis-Hastings and Gibbs

The Algorithms That Build the Chain

Propose and accept · the evidence cancels · tuning · Gibbs sampling

Part 1 said: build a chain whose stationary law is the posterior. Here is how. Metropolis-Hastings proposes moves and accepts them by a simple ratio; Gibbs draws one parameter at a time.

2

PART 2 · THE ALGORITHMS

Building the Chain

The Metropolis-Hastings accept-reject rule, why the intractable evidence cancels, tuning the proposal step, and Gibbs sampling as the conditional-by-conditional alternative.

From Idea to Algorithm

Part 1 gave the plan; Part 2 builds the chain

1

We want posterior samples

Part 1: build a Markov chain whose stationary distribution is the posterior, then read answers off the draws.

2

But how do we build such a chain?

We need a concrete rule for choosing the next state that guarantees the right long-run distribution.

3

Two classic recipes do it

Metropolis-Hastings proposes and accepts moves; Gibbs updates one parameter at a time. Both give a valid chain.

Part 2 in one line: propose-and-accept (Metropolis-Hastings) or draw-conditionals (Gibbs) build the chain.

The Metropolis-Hastings Algorithm

Propose a move, then decide whether to take it

Metropolis-Hastings is disarmingly simple: from the current point, propose a nearby candidate. If it makes the target higher, take it; if lower, take it only sometimes. Repeat, and the chain traces out the posterior.

The loop

- Propose a candidate near the current point.
- Compare the target at the candidate and the current point.
- Accept or reject by a simple probability rule.

Why it is remarkable

- It needs only the target's value at two points.
- No integral, no normalizer, no special structure.
- It works for an enormous range of models.

The Acceptance Rule

One ratio decides every move

acceptance: $\alpha = \min\left(1, \frac{p(\theta^*|\text{data})}{p(\theta|\text{data})}\right)$ accept θ^* with prob. α

Reading it

- Form the ratio of the target at candidate over current.
- If the candidate is better, the ratio exceeds one — always accept.
- If worse, accept it with probability equal to the ratio.

The effect

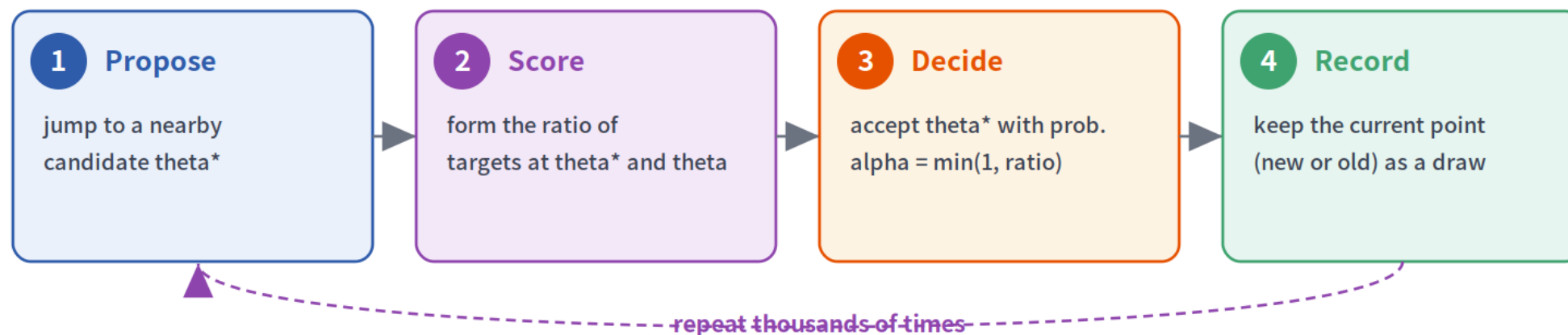
- The chain climbs toward high-probability regions.
- But it still wanders into lower ones sometimes.
- That balance is what maps the whole distribution.

One Metropolis-Hastings Step

Propose a candidate, score it by the ratio of targets, accept with the resulting probability, and record the point. Repeat thousands of times, and the recorded draws reproduce the posterior.

One Metropolis-Hastings Step

Propose a move, score it by a ratio, then accept or reject - and repeat.



Why it works

Uphill moves are always accepted; downhill moves sometimes are. That balance makes the chain linger where the target is high - so its long-run visits reproduce the posterior. The normalizer never appears.

The Trick: The Evidence Cancels

Why we never need the hard integral

Here is the magic. The acceptance rule uses a RATIO of posteriors. The intractable normalizer appears in both the top and the bottom — so it cancels. We only ever need the likelihood times the prior.

What cancels

- The posterior is likelihood times prior, over the evidence.
- In the ratio, the evidence is in both numerator and denominator.
- So it divides out completely.

Why this is everything

- The evidence was the whole obstacle (Week 13).
- MCMC never has to compute it.
- That is why the method works where exact Bayes fails.

The Evidence Divides Out

Seeing it happen

$$\frac{p(\theta^*|\text{data})}{p(\theta|\text{data})} = \frac{P(\text{data}|\theta^*)P(\theta^*)/P(\text{data})}{P(\text{data}|\theta)P(\theta)/P(\text{data})} \Rightarrow P(\text{data}) \text{ cancels}$$

The two posteriors

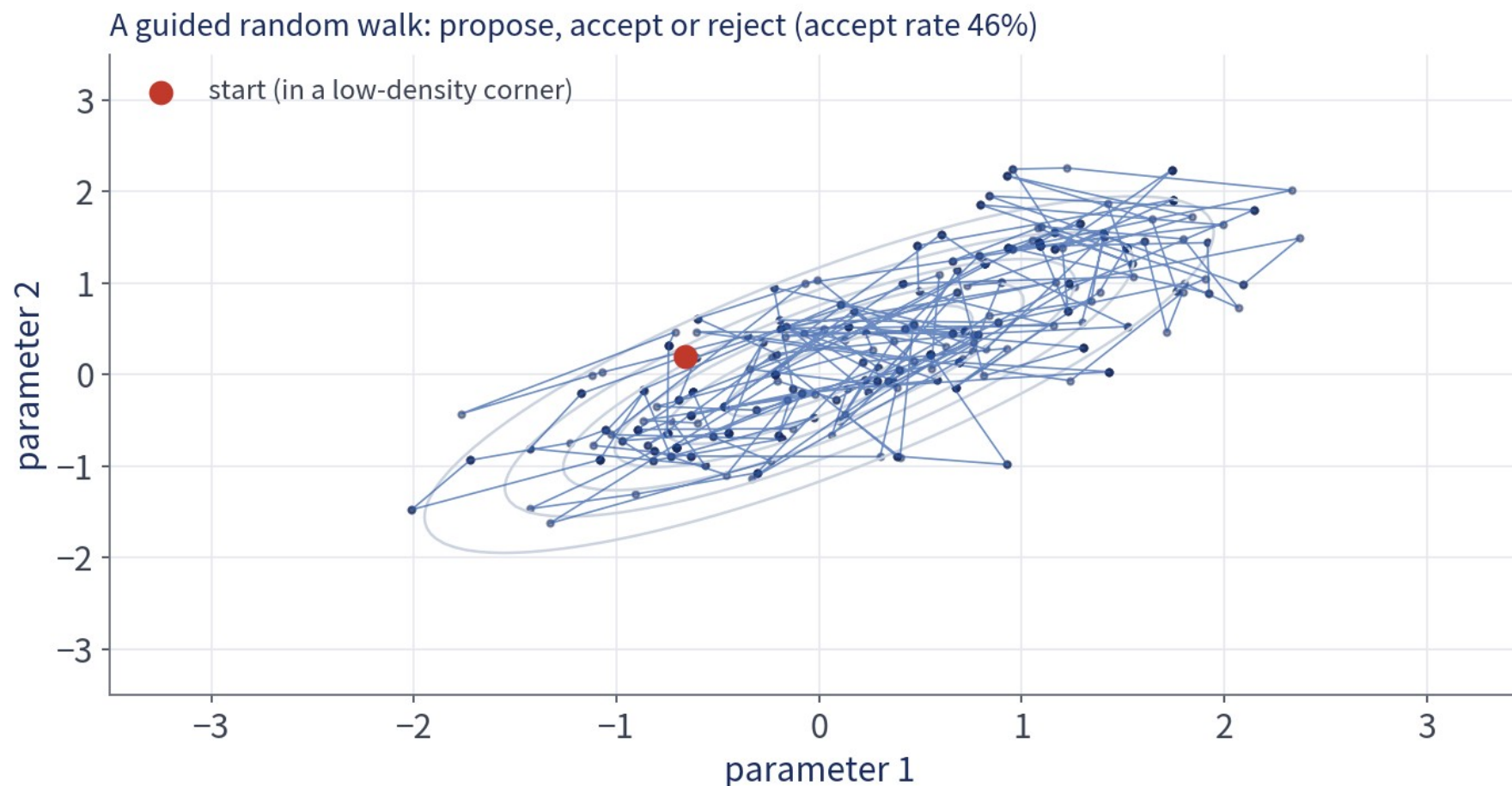
- Both are likelihood times prior, over the same evidence.
- That evidence is the identical constant in each.
- It does not depend on the parameter.

The result

- The constant cancels in the ratio.
- Only likelihood times prior remains — computable.
- The hard integral simply never appears.

A Guided Random Walk

Starting in a low-density corner, the chain proposes moves and accepts or rejects them, quickly finding and then exploring the high-probability region traced by the target's contours.



Why Accept Some Worse Moves?

Exploration versus exploitation

If we only ever climbed

- The chain would rush to the peak and stop.
- It would sample one point, not the distribution.
- We would learn the mode, not the uncertainty.

By sometimes stepping down

- The chain explores the full spread of the posterior.
- It visits the tails in the right proportion.
- That is what makes the draws represent the whole shape.

The Proposal Distribution

How candidates are generated

What it is

- The rule for proposing a candidate from the current point.
- Often a Gaussian step of some chosen width.
- It is the one thing you must design.

Why it matters

- It sets how far the chain tries to move each step.
- Too timid or too bold both hurt (next slide).
- Getting it right is most of the practical art.

Tuning the Proposal Step

Too small, too big, or just right

The proposal width is a dial you must tune. Too small and the chain inches along, exploring painfully slowly. Too big and almost every move is rejected, so it gets stuck. The sweet spot moves freely AND accepts often enough.

The two failure modes

- Too small: high acceptance, but tiny steps — slow.
- Too big: bold jumps, but constant rejection — stuck.
- Both explore the posterior badly.

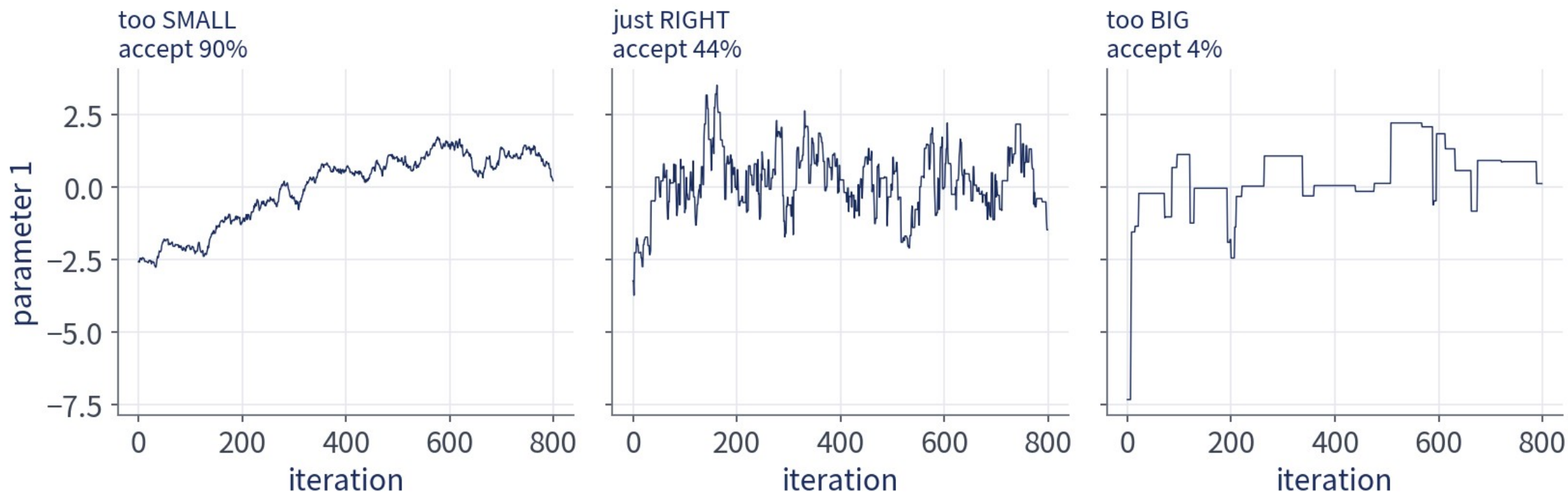
The sweet spot

- Steps large enough to move across the distribution.
- Small enough that many are accepted.
- A moderate acceptance rate is the sign of good tuning.

Too Small, Just Right, Too Big

Three chains, differing only in proposal width. A tiny step crawls and barely moves; a huge step gets stuck rejecting; a well-chosen step mixes freely across the distribution.

Tuning the proposal step: too small crawls, too big gets stuck, just right mixes



The Target Acceptance Rate

A dial you can read

What to watch

- The fraction of proposals the chain accepts.
- Very high means the steps are too timid.
- Very low means the steps are too bold.

The guidance

- For random-walk moves, aim for a moderate rate.
- Roughly a quarter to a half is a common target.
- Modern samplers tune this automatically.

Gibbs Sampling

Update one parameter at a time

Gibbs sampling takes a different route. Instead of proposing joint moves, it cycles through the parameters, drawing each from its distribution given all the others. No proposal, no rejection — every draw is accepted.

The idea

- Fix all parameters but one.
- Draw that one from its full conditional distribution.
- Cycle through them all, then repeat.

The appeal

- No proposal width to tune.
- No rejected moves — every step advances.
- Clean and efficient when the conditionals are known.

The Gibbs Update

Draw each conditional in turn

Gibbs: draw each in turn $\theta_j \sim p(\theta_j \mid \theta_{-j}, \text{data})$

Reading it

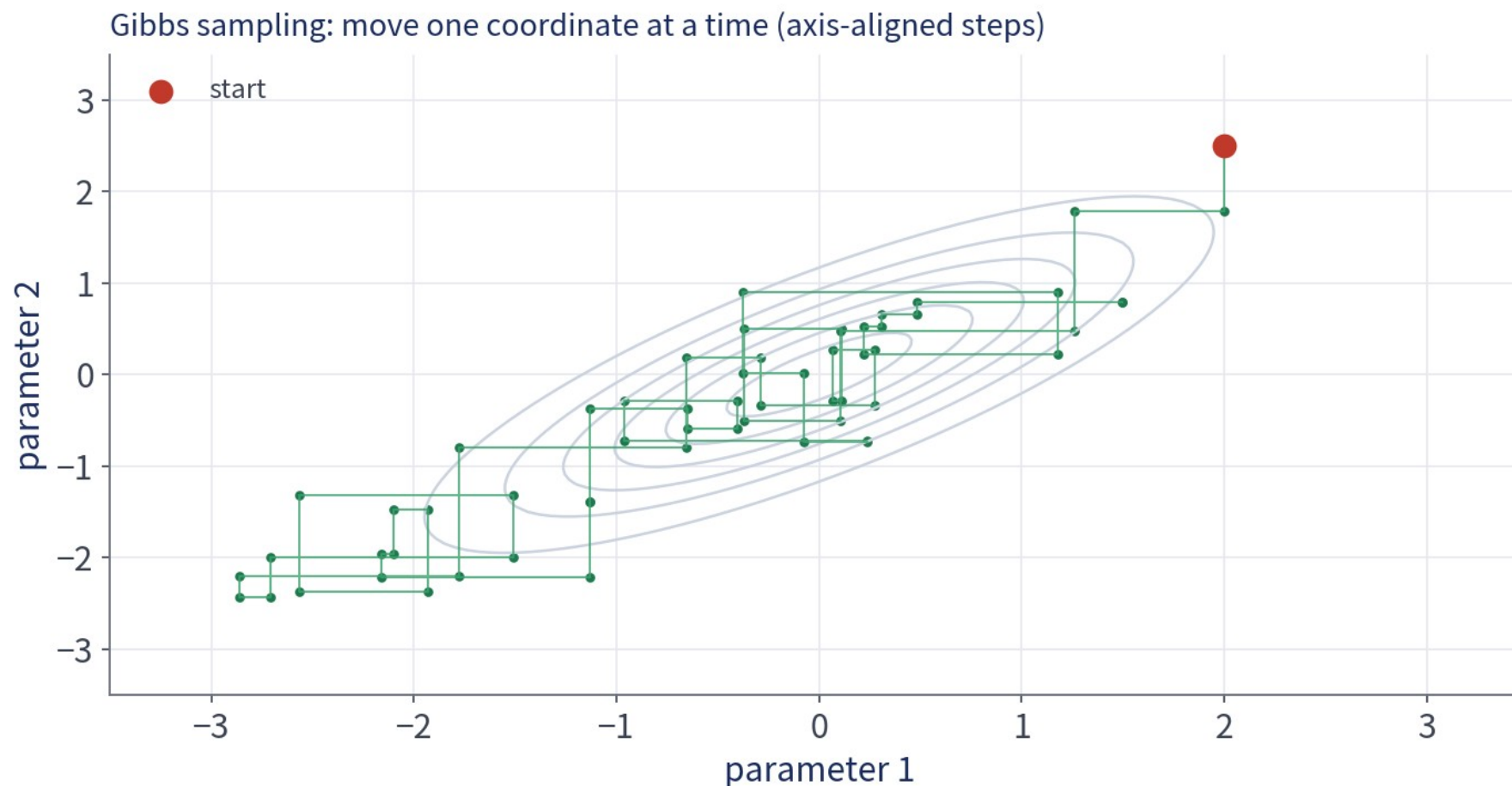
- For each parameter, condition on all the others.
- Draw it from that full conditional distribution.
- Sweep through every parameter, then loop.

The requirement

- You need each full conditional in closed form.
- Conjugate models often provide exactly this.
- When you have them, Gibbs is very clean.

Gibbs Moves One Coordinate at a Time

Each step is axis-aligned: hold one parameter, draw the other from its conditional. The zig-zag path fills the same target, but its stair-step motion can be slow when parameters are correlated.



Two Ways to Build the Same Chain

Metropolis-Hastings proposes and rejects and works almost anywhere, but must be tuned. Gibbs draws exact conditionals with no tuning or rejection, but needs those conditionals and struggles with correlation

Two Ways to Build the Same Chain

Metropolis-Hastings proposes and rejects; Gibbs draws exact conditionals, one at a time.

Metropolis-Hastings

- Propose a move, accept or reject.
- Needs only the target, up to a constant.
- Works for almost any model.
- Must tune the proposal step.
- Rejections waste some proposals.

General-purpose workhorse.

Gibbs sampling

- Update one parameter at a time.
- Draw each from its full conditional.
- No tuning, no rejections.
- Needs those conditionals in closed form.
- Struggles when parameters are correlated.

Elegant when conditionals are known.

When to Use Which

Matching sampler to model

Reach for Metropolis-Hastings

- The model is arbitrary or non-conjugate.
- You have no closed-form conditionals.
- You want one general-purpose tool.

Reach for Gibbs

- The full conditionals are known and easy to sample.
- The model is conjugate in its pieces.
- You want to avoid tuning and rejections.

Why the Chain Has the Right Target

The balance condition, in words

The guarantee

- The accept-reject rule is built to satisfy a balance condition.
- It makes flows between states cancel at the posterior.
- That forces the posterior to be the stationary distribution.

What it means for you

- You don't have to prove this each time.
- The standard rules come with the guarantee.
- Run the chain, and it targets the posterior by design.

Both Build the Same Kind of Chain

Different moves, same destination

What they share

- Both produce a Markov chain over the parameters.
- Both have the posterior as their stationary law.
- Both yield correlated-but-valid posterior draws.

How they differ

- MH proposes joint moves and may reject.
- Gibbs updates coordinates and always accepts.
- Gibbs is in fact a special case of the MH family.

What Part 3 Adds

From a running chain to trustworthy answers

Diagnosing the chain

- Trace plots: is it mixing well?
- Burn-in: discard the awkward start.
- Convergence checks before you trust the draws.

Doing it for real

- Modern tools like PyMC that sample and tune for you.
- Fitting the messy models from earlier weeks.
- The bridge to Bayesian state-space models.

Code a Metropolis Sampler by Hand

Direct Claude Code to demystify the algorithm

GOAL Build a Metropolis sampler from scratch and watch it recover a target distribution.

[INPUT] Prompt · Claude Code

Write and run metropolis.py that:

1. defines a target as an UNNORMALIZED density (e.g. a mixture of two Gaussians);
 2. runs a random-walk Metropolis sampler using only the accept ratio $\min(1, \text{target}^*/\text{target})$;
 3. tries three proposal widths (too small, good, too big) and reports each acceptance rate;
 4. overlays the sampled histogram on the target.
- Confirm the histogram matches the target and note which proposal width mixes best.

How to work with it

- Only the target ratio is needed.
- The normalizer never appears.
- Tune the width to a moderate accept rate.
- The histogram should match the target.

Deliverable: metropolis.py + sampled histograms at three step sizes with acceptance rates.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 In the MH acceptance rule, what makes the normalizer cancel?

It's a ratio of posteriors

Q2 Why does MH sometimes accept a worse move?

To explore the whole posterior

Q3 What does Gibbs sampling draw each parameter from?

Its full conditional

All three come from the acceptance ratio, exploration, and the Gibbs conditional draws.

Three Misconceptions to Avoid

Where sampler intuition slips

Common myths

- "A higher acceptance rate is always better."
- "Gibbs is a totally different method from MH."
- "Rejected proposals are wasted and skipped."

The reality

- Too-high acceptance means tiny, slow steps — tune it down.
- Gibbs is a special case of the MH family.
- On rejection you re-record the CURRENT point as a draw.

Three Things to Carry Forward

Part 2 in one screen

1

Metropolis-Hastings proposes and accepts

From the current point, propose a candidate and accept it by a simple ratio of targets. Uphill moves are always taken; downhill moves sometimes are, so the chain maps the whole posterior.

2

The evidence cancels

Because the acceptance rule is a ratio of posteriors, the intractable normalizer divides out. MCMC needs only the likelihood times the prior — which is why it works where exact Bayes fails.

3

Gibbs is the conditional-by-conditional alternative

It draws each parameter from its full conditional in turn — no tuning, no rejection — when those conditionals are available. Both methods build a chain with the posterior as its stationary law.

Next — Part 3: trace plots, burn-in, convergence diagnostics, and MCMC in practice with PyMC.